



铁头山羊STM32教程

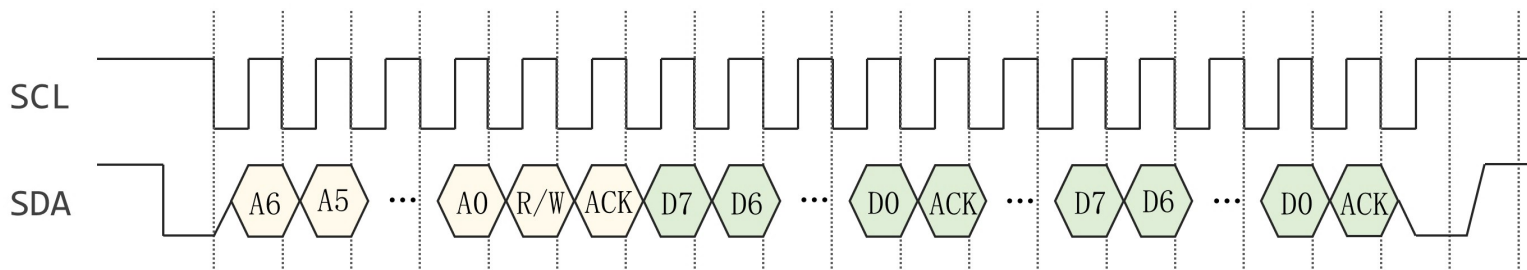
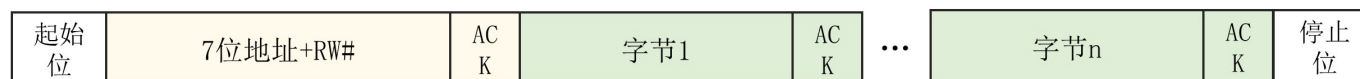
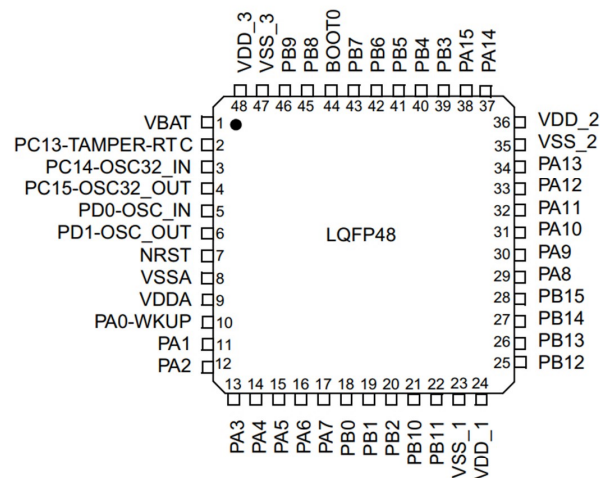
[I2C]

4. 6. 软I2C

版权归东莞市明玉科技有限公司所有
未经允许不得传播

1. 操作繁琐

2. 引脚位置受到限制



版权归东莞市明玉科技有限公司所有

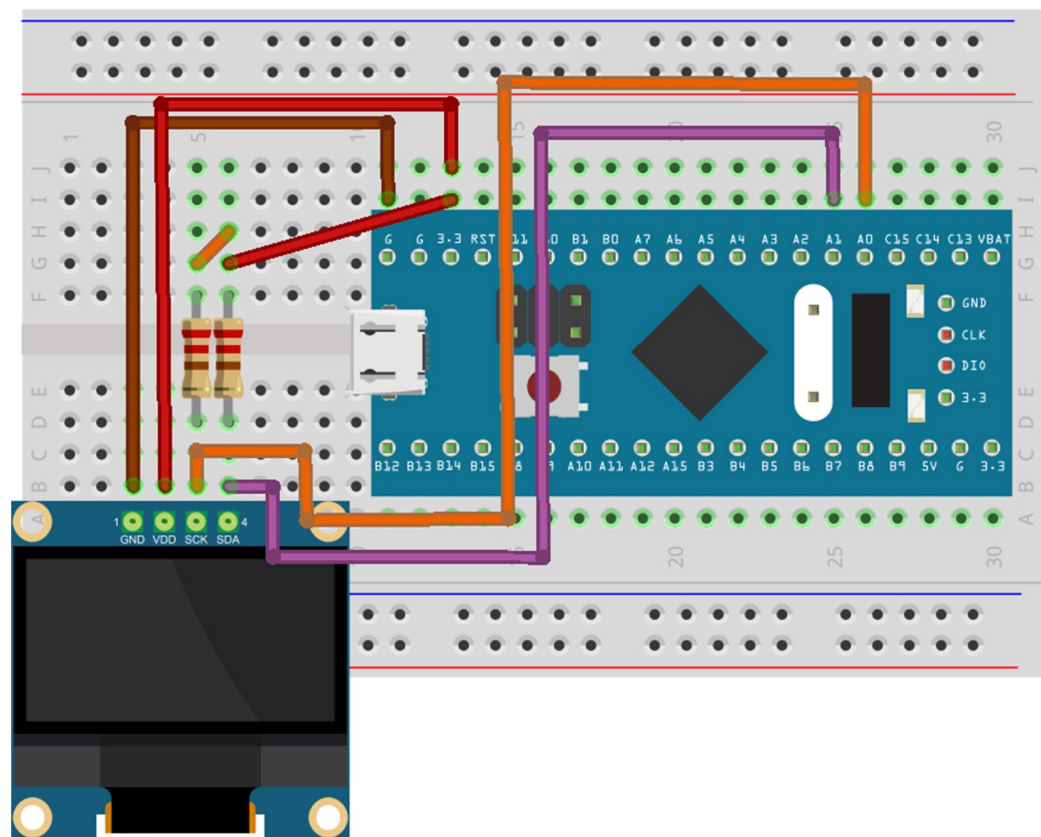
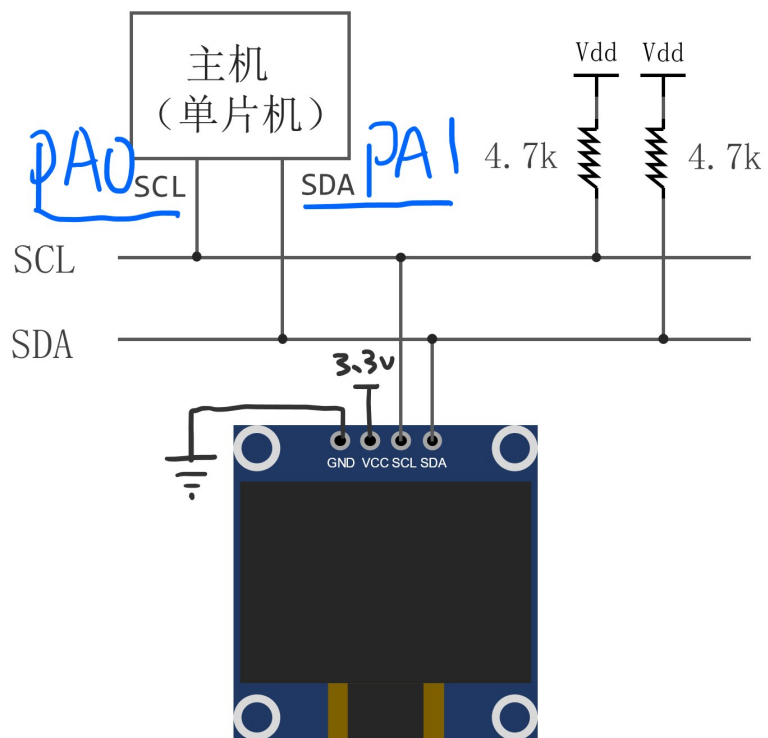
未经允许不得传播

- 1. IO引脚初始化
- 2. IO读写和延迟函数
- 3. 发送起始位和停止位
- 4. 发送一个字节
- 5. 接收一个字节
- 6. 综合
- 7. 测试

版权归东莞市明玉科技有限公司所有

未经允许不得传播

通用输出开漏(初始电压高)



版权归东莞市明玉科技有限公司所有

未经允许不得传播

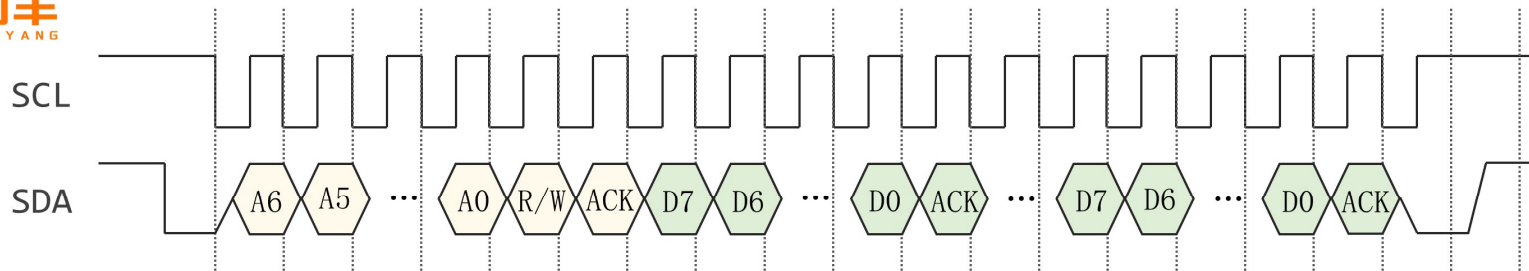


百度网盘地址

<https://pan.baidu.com/s/1kcKd063anDtFZ1gG97cOKg?pwd=3kvt> 提取码: 3kvt



版权归东莞市明玉科技有限公司所有
未经允许不得传播



```
void scl_write(uint8_t level); // 向SCL写0或者写1
```

```
void sda_write(uint8_t level); // 向SDA写0或者写1
```

```
uint8_t sda_read(void); // 读取SDA的值
```

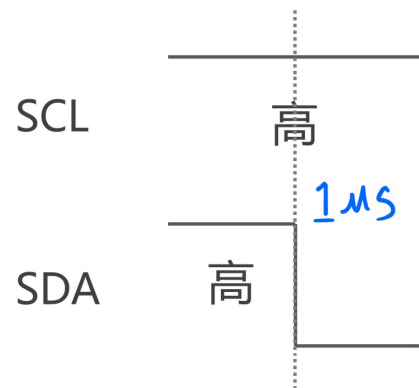
```
void delay_us(uint32_t us) // 微秒级延迟
```

```
{  
    uint32_t n = us*8;  
    for(uint32_t i=0; i<n; i++);  
}
```

版权归东莞市明玉科技有限公司所有

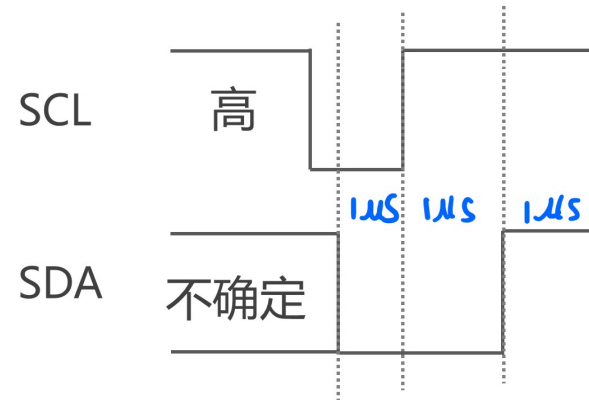
非经允许不得传播

起始位



```
sda_write(0);  
delay(1);
```

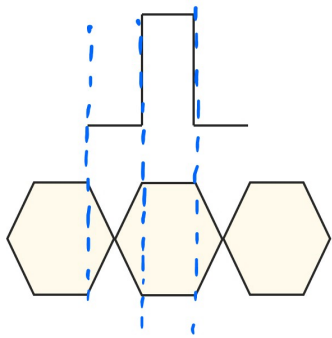
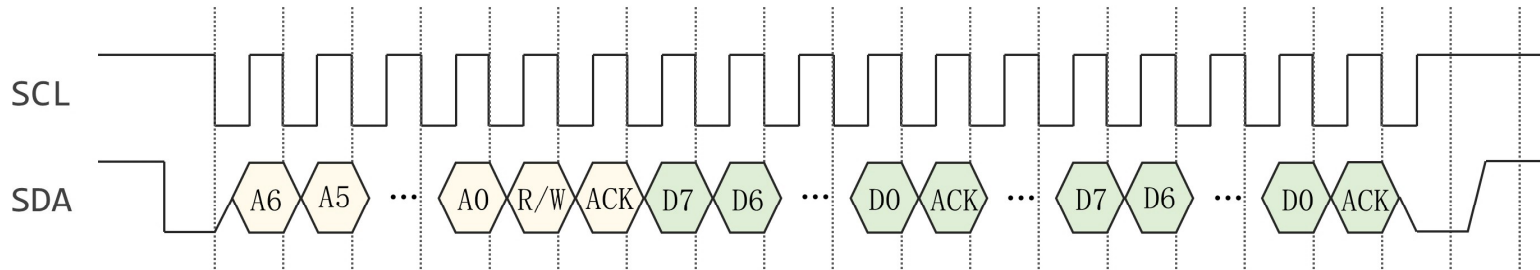
停止位



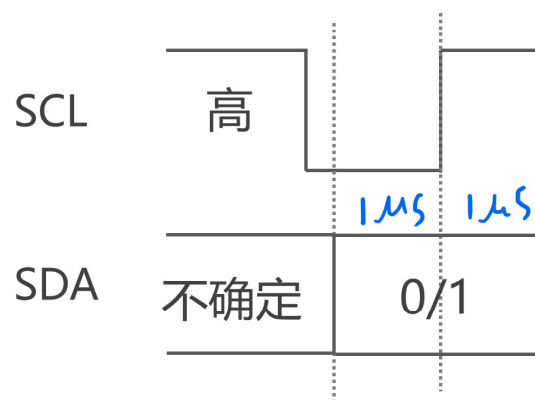
```
scl_write(0);  
sda_write(0);  
delay(1);  
scl_wirte(1);  
delay(1);  
sda_write(1);  
delay(1);
```

版权归东莞市明玉科技有限公司所有

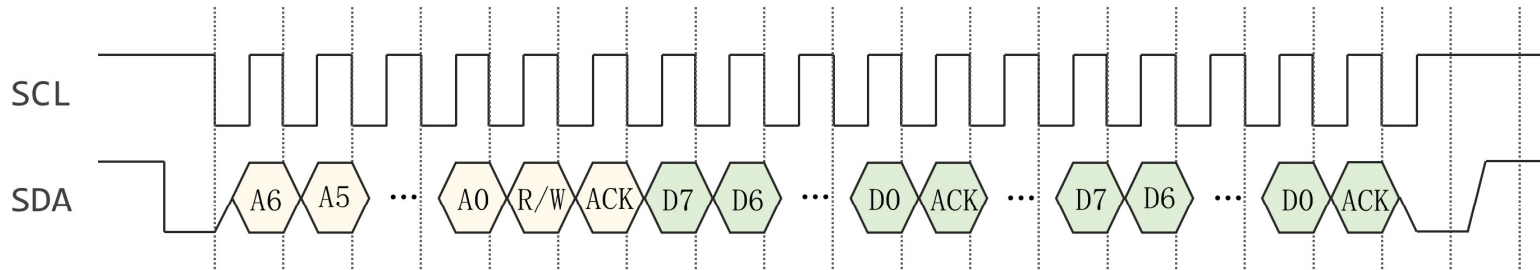
未经允许不得传播



scl=0, 发送方准备数据
scl=1, 接收方读取数据

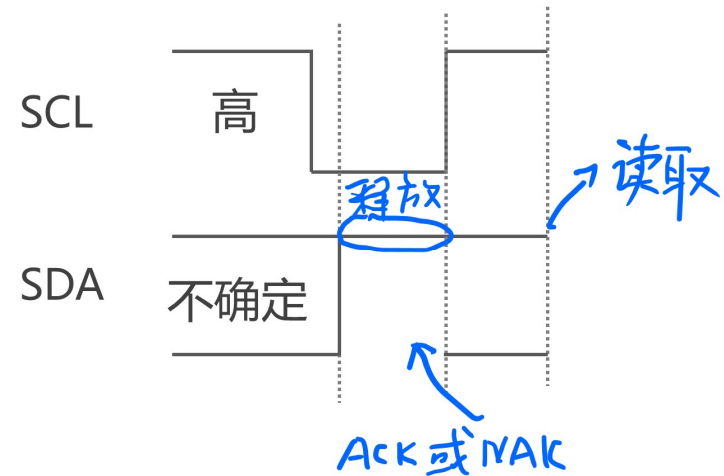


```
scl_write(0);
sda_write(x);
delay(1);
scl_wirte(1);
delay(1);
```

`uint8_t send_byte(uint8_t byte); // 发送一个字节, 返回0表示ACK`

```
for(int8_t i=7; i>=0; i--){
    scl_write(0);sda_write(x);delay(1);
    scl_write(1);delay(1);
}
// 读取ACK或NAK
scl_write(0); sda_write(1); delay(1);
scl_write(1); delay(1);
return sda_read();
```



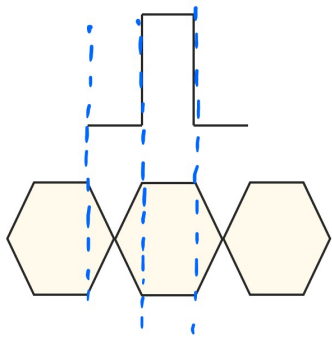
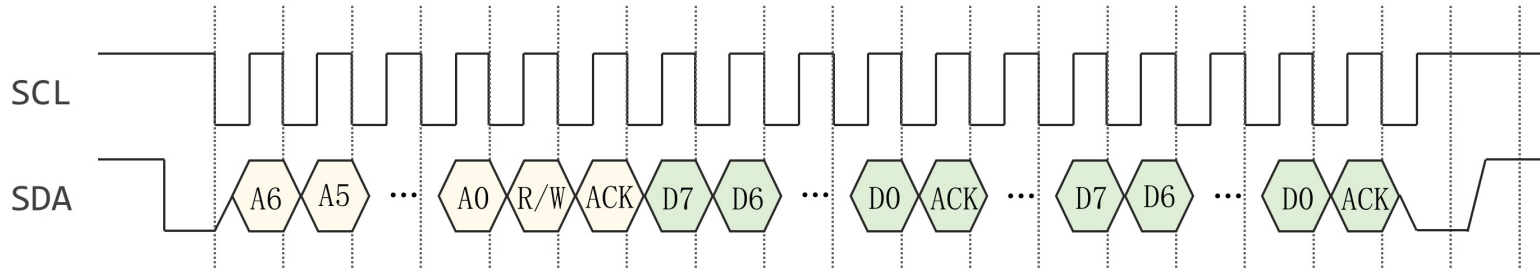
版权归东莞市明玉科技有限公司所有
未经允许不得传播



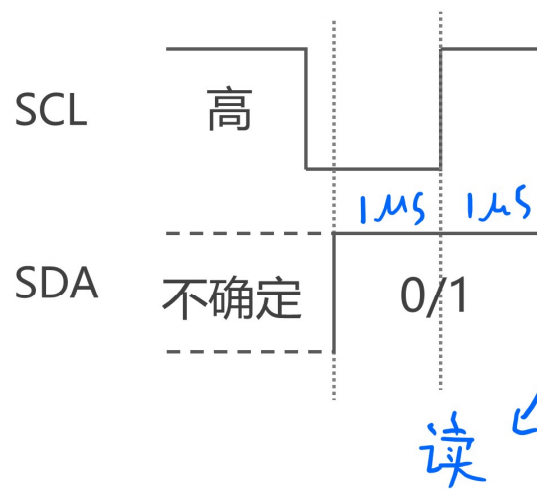
```
if(Byte & (0x01 << i) != 0){  
    // 第i位等于1  
}  
else{  
    // 第i位等于0  
}
```

Byte	D7	D6	D5	D4	D3	D2	D1	D0
& 0x01 << 3	0	0	0	0	1	0	0	0
	0	0	0	0	D3	0	0	0

版权归东莞市明玉科技有限公司所有
未经允许不得传播

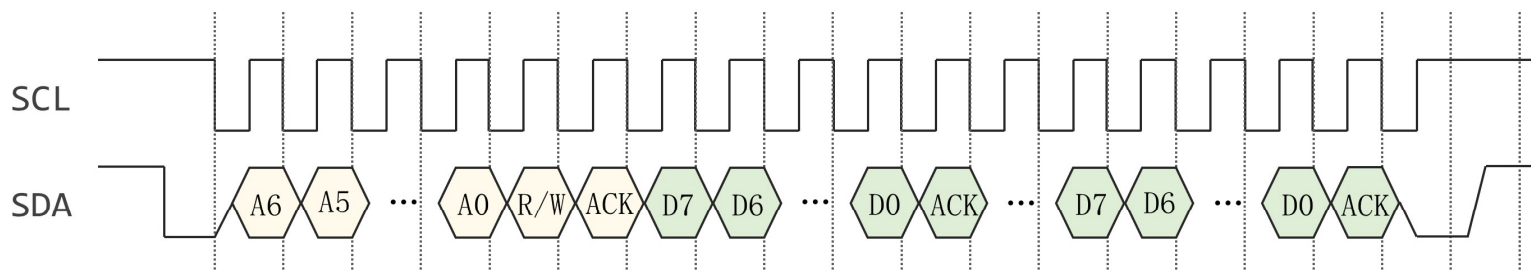


scl=0, 发送方准备数据
scl=1, 接收方读取数据



```
scl_write(0);
sda_write(1); // 释放SDA
delay(1);
scl_wirte(1);
delay(1);
bit = sda_read();
```

版权归东莞市明玉科技有限公司所有
未经允许不得传播

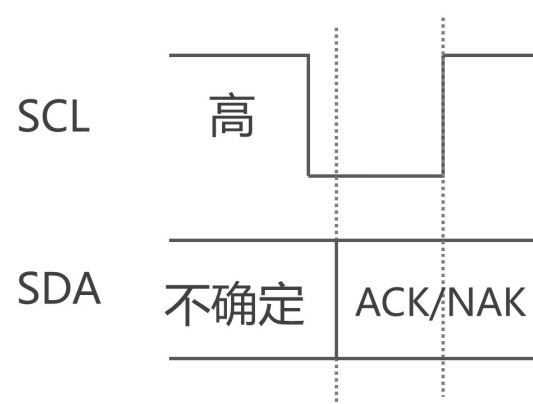


```
uint8_t receive_byte(uint8_t Ack); // 读取一个字节
```

```
uint8_t byte = 0;
for(int8_t i=7; i>0; i--){
    scl_write(0); sda_write(1); delay(1);
    scl_write(1); delay(1);
    if(sda_read()!=0)
        byte |= 0x01 << i;
}
```

// 回复ACK或NAK

```
scl_write(0); sda_write(!Ack); delay(1);
scl_write(1); delay(1); return byte;
```



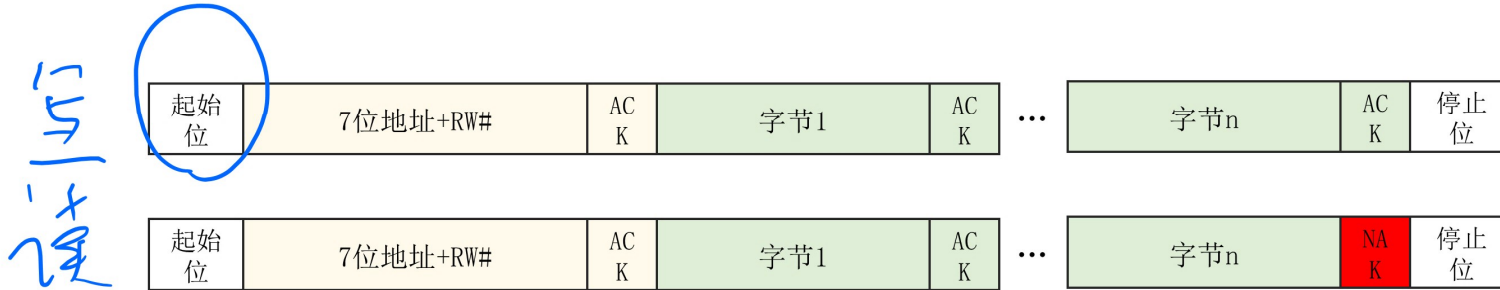
版权归东莞市明玉科技有限公司所有
未经允许不得传播

软I2C写

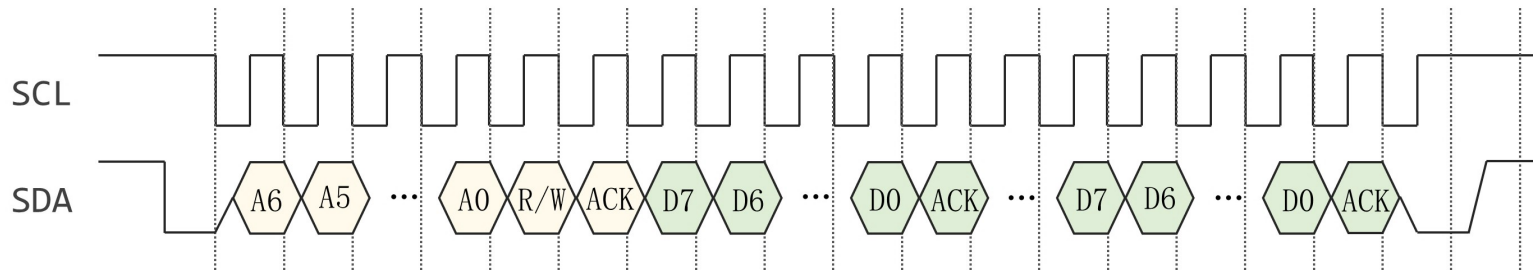
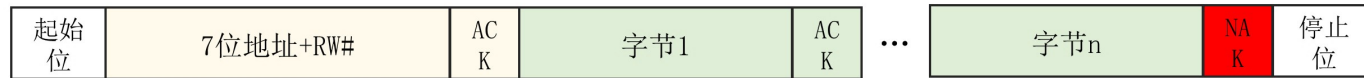
```
int My_SI2C_SendBytes(uint8_t Addr, // 从机地址, 靠左
                      uint8_t *pData, // 要发送的数据
                      uint16_t Size); // 要发送的数据的数量 (字节)
```

软I2C读

```
int My_SI2C_ReceiveBytes(uint8_t Addr, // 从机地址, 靠左
                         uint8_t *pBuffer, // 接收缓冲区
                         uint16_t Size); // 要发送的数据的数量 (字节)
```



版权归东莞市明玉科技有限公司所有
未经允许不得传播



```
int My_I2C_SendBytes(...){
    SendStart(); // 发送起始位
    if(SendByte(Addr&0xfe)!=0){ // 发送地址
        SendStop(); return -1; // 寻址失败
    }
    for(uint16_t i=0; i<Size; i++){
        if(SendByte(pData[i])!=0){ //发送数据
            SendStop(); return -2; // 拒收
        }
    }
    SendStop(); return 0; } // 发送停止位
```

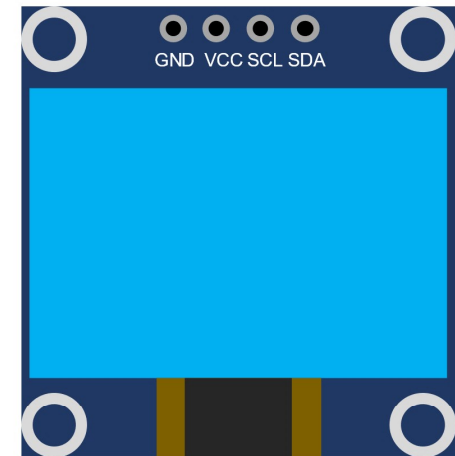
```
int My_I2C_ReceiveBytes(...){
    SendStart(); // 发送起始位
    if(SendByte(Addr|0x01)!=0){ // 发送地址
        SendStop(); return -1; // 寻址失败
    }
    for(uint16_t i=0; i<Size-1; i++){
        pBuffer[i] = ReceiveByte(1);
    }
    pBuffer[Size-1] = ReceiveByte(0);
    SendStop(); return 0; } // 发送停止位
```

版权归东莞市明志科技有限公司所有

未经允许不得传播

0x78

```
uint8_t commands[] = {  
    0x00, // 命令流  
    0x8d, 0x14, // 使能电荷泵  
    0xaf, // 打开屏幕开关  
    0xa5, // 让屏幕全亮  
}
```



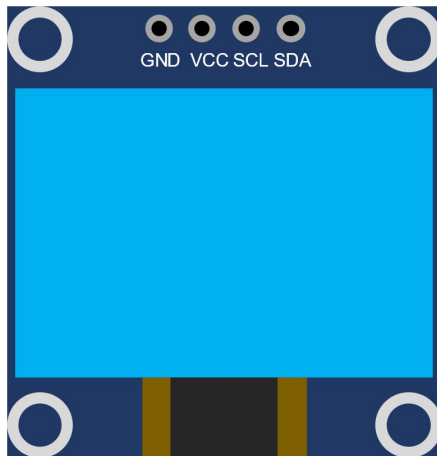
版权归东莞市明玉科技有限公司所有
未经允许不得传播

Table 9-2 : Read Command Table

Bit Pattern	Command	Description
D ₇ D ₆ D ₅ D ₄ D ₃ D ₂ D ₁ D ₀	Status Register Read	D[7]: Reserved D[6]: “1” for display OFF / “0” for display ON D[5]: Reserved D[4]: Reserved D[3]: Reserved D[2]: Reserved D[1]: Reserved D[0]: Reserved

$D_6 = 0$, 表示屏幕开启

$D_6 = 1$, 表示屏幕关闭



```
uint8_t rcvd;  
My_I2C_ReceiveBytes(I2C1, 0x78,&rcvd,1);  
if((rcvd & (0x01<<6))==0){ // 检测到屏亮  
    GPIO_WriteBit(GPIOC, GPIO_Pin_13, Bit_RESET); // 亮  
}else{ // 检测到屏灭  
    GPIO_WriteBit(GPIOC, GPIO_Pin_13, Bit_SET); // 灭灯  
}
```

版权归东莞市明玉科技有限公司所有

未经允许不得传播